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Comment

The forgotten creator: Towards a statutory remuneration right for machine learning of generative AI

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ABSTRACT

Generative AI is disrupting the creative process(es) of intellectual works on an unparalleled scale. Algorithmic tools are increasing users' production capacity for literary and artistic works to almost infinite levels. However, the quality of the outputs is strictly dependent on the quantity and quality of the inputs, some of which are protected by copyright. This scenario gave raise to tensions between copyright holders and generative AI companies. While the formers claim control over this new kind of exploitation of their works, the latters wish to train their algorithms freely with as many contents as possible. This contribution suggests exploring the idea of introducing a statutory license for machine learning purposes as a compromise solution to ensure an attractive environment for artificial intelligence without marginalizing the role played by human authors. This remuneration proposal is rooted in a fundamental rights analysis that balances i.e., the right to science and culture and freedom of artistic expression (Arts. 11 and 13 EUCF, 19 UDHR, 27.1 UDHR, 15.1 a and b ICESCR) vis-à-vis the right for creators to benefit from the protection of the moral and material interests resulting from their scientific, literary or artistic production (Arts. 17.2 EUCF, 27.2 UDHR, and 15.1 c ICESCR).

1. The growing frictions of generative AI with copyright law and the fundamental rights at stake

Generative Artificial Intelligence ('AI') is disrupting the creative process(es) of intellectual works on an unparalleled scale. More and more AI systems offer services that push users' production capacity of new literary and artistic works beyond unforeseen barriers. Algorithmic tools are gradually colonizing every creative sector, from being able to generate text (i.e., ChatGPT,[3] Smodin[4]), to perform music (i.e., AIVA,[5] Beatoven,[6] Soundful[7]), to draw images (i.e., Dall-e,[8] Midjourney,[9]

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[3] * While the ideas presented in this paper are commonly shared by the two authors, paragraphs 1 and 2 are written by Vincenzo Iaia and paragraphs 3 and 4 are written by Christophe Geiger.The service is available at the following link: https://chat.openai.com/

[4] https://smodin.io/it

[5] https://www.aiva.ai/

[6] https://www.beatoven.ai/

[7] https://soundful.com/

[8] https://openai.com/dall-e-2

[9] https://www.imagine.art/?utm_source=google&utm_medium=cpc&utm_campaign=Competitor_web_Tier2&gclid=Cj0KCQjw0IGnBhDUARIsAMwFDLmsQpiDysWhy1jTOYoof2m5fWtyVMF0psLT_9TXH0dN_mFCQtDZdUcaAvpeEALw_wcB

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DreamStudio[10]), and to shoot movies (i.e., Deepbrain AI,[11] Veed.io[12]). It is estimated that the global Generative-AI market will sharply grow in the next decade, from $ 10.6 billion 2022 to $ 200.7 billion by 2032.[13] Apart from redefining the traditional business models of creative industries, the ability to obtain new artworks with an increasingly marginalization of human contribution – sometimes reduced to a handful of prompts – has inevitably tested the fitness of copyright legislations in all over the world to deal with the so-called "artificial intelligence" revolution.

In a nutshell, Generative AI raises two main copyright issues[14] that branch off into further sub-problems which in turn intercept (if not collide with) some fundamental rights, especially freedom of artistic expression, freedom of art and science and the right to science and culture (Arts. 11 and 13 EUCFR,[15] 19 UDHR,[16] 27.1 UDHR, Art. 15.1 a and b ICESCR[17]) and the right to the protection of moral and material interests of creators (Arts. 17.2 EUCFR, 27.2 UDHR, and 15.1 c ICESCR).

From the input side, it is questionable whether the training of AI through the extraction and mining of copyrighted works constitutes a copyright infringement or falls within an exceptional regime which varies from Europe to the United States and other parts of the world[18] (in particular, Japan has an interesting copyright limitation which might apply[19]). Indeed, human creators seek compensation for the novel use of their intellectual efforts while AI firms aim to maximize the free harvesting of data (including copyright-protected materials) for training their algorithms. From the output side, it is hotly debated whether the content produced by a generative AI satisfies the protectability requirement under copyright law to trigger the exclusive protection.[20]

Courts are already dealing with the first question since some content creators and licensees have filed copyright infringement lawsuits against providers of generative-AI services (namely Open AI, Meta, Stability AI, Anthropic, and Midjourney).[21] The Northern District Court of California rendered a first decision in which most of the artists' claims against Stability AI and Midjourney were predominantly dismissed for lack of evidence.[22] However, the dispute may not be conclusively resolved since the losing parties can amend and file their complaints again to clarify their theories on how each defendant violated their copyrights. These litigations might have convinced the European legislator to deal with the issue in the proposal for a regulation laying down harmonized rules on Artificial Intelligence ('Artificial Intelligence Act' or 'AIA'), introducing lately a provision to address the issue of transparency with regard to the works used in the machine learning process.[23] Very recently, IP offices

[10] https://beta.dreamstudio.ai/generate

[11] https://www.deepbrain.io/aistudios?via=antoine

[12] https://www.veed.io/

[13] OECD, 'Initial Policy Considerations for Generative Artificial Intelligence', OECD Artificial Intelligence Papers, No. 1, 2023, 26, mentioning the report by Polaris, 'Generative AI Market Share, Size, Trends, Industry Analysis Report; Segment Forecast 2023-2032', 2023, available at https://www.polarismarketresearch.com/industry-analysis/generative-ai-market.

[14] For a structured review of the issues arising from the mushrooming intersections between AI and Intellectual Property see Daniel Gervais 'Checklist of Issues on Generative IP' Kluwer Copyright Blog, 11 September 2023, available at: https://copyrightblog.kluweriplaw.com/2023/09/11/checklist-of-issues-on-generative-ip/.

[15] Charter of Fundamental Rights of the European Union, 26 October 2012, C326/391.

[16] Universal Declaration of Human Rights, 10 December 1948, General Assembly Resolution 217 A.

[17] International Covenant on Economic, Social and Cultural Rights, 16 December 1966, General Assembly resolution 2200 A.

[18] On this topic see, among many scholars, Christophe Geiger, 'The missing goal-scorers in the artificial intelligence team: of big data, the fundamental right to research and the failed text and data mining limitations in the CDSM Directive' in Martin Senftleben, Joost Poort, Mirelle van Eechoud, Stef van Gompel, Natali Helberger (eds.), Intellectual Property and Sports, Essays in Honour of P. Bernt Hugenholtz (Kluwer International, 2021) 383-394; Thomas Margoni & Martin Kretschmer, 'A deeper look into the EU text and data mining exceptions: harmonization, data ownership, and the future of technology' (2022) 71(8) GRUR International 685-701.

[19] Tatsuhiro Ueno, 'The Flexible Copyright Exceptions for 'Non-Enjoyment' Purposes – Recent Amendment in Japan and Its Implication' (2021) 70(2) GRUR International 145-152; Artha Dermawan, 'Text and data mining exceptions in the development of generative AI models: What the EU member states could learn from the Japanese "nonenjoyment" purposes?' (2023) The Journal of World Intellectual Property 1-25.

[20] On the terms of the debate see the study conducted by Oleksandr Bulayenko, Joao Pedro Quintais, Daniel Gervais and Joost Poort, 'AI Music Outputs: Challenges to the Copyright Legal Framework', reCreating Europe Report, 2022; Bernt Hugenholtz and Joao Pedro Quintais, 'Copyright and Artificial Creation: Does EU Copyright Law Protect AI-Assisted Output?' (2021) 52 International Review of Intellectual Property and Competition Law 1190 sq.

[21] In January, three artists sued Open AI and Meta before the US District Court for the Northern District of California on the grounds that their algorithms were trained by extracting (also) their works without clearing any license. For similar reasons Getty Images sued Stability AI Inc. alleging that the company exploited 12 million images to train its AI model. For more information see respectively Riddhi Setty, 'Sarah Silverman, Authors Hit OpenAI, Meta With Copyright Suits' Bloomberg Law, 10 July 2023, available at: https://news.bloomberglaw.com/ip-law/sarah-silverman-authors-hit-openai-meta-with-copyright-suits; Blake Brittain, 'Getty Images lawsuit says Stability AI misused photos to train AI' Reuters, 6 February 2023, available at: https://www.reuters.com/legal/getty-images-lawsuit-says-stability-ai-misused-photos-train-ai-2023-02-06/. On 19 September 2023, the Authors Guild and several (famous) professional writers brought a class action against OpenAI before the Southern District Court of New York, No. 1:23-cv-8292. The plaintiffs seek compensation for copyright infringement arguing that their works of fiction have been knowingly copied without their consent for training the large language models that enable OpenAI to offer its commercial Generative AI-based services, such as ChatGPT Plus and ChatGPT Enterprise. Also, on 18 October 2023, Universal Music and various music publishers sued the company Anthropic, a provider of generative AI services, before the District Court for the Middle District of Tennessee, No. 3: 23-cv-01092. The claimants argue that their copyright-protected musical compositions are scraped and ingested into the defendant's AI models without securing permission. To this purpose, the music publishers provided proof that the AI system offered by Anthropic, when prompted with songs by Katy Perry, Gloria Gaynor, Garth Brooks, and the Rolling Stones, reproduced substantial parts of their lyrics. More generally, the use of AI in the creative industries led Hollywood screenwriters to go on strike for obtaining a regulatory intervention able to protect their intellectual works as well to avoid their replacement by AI tools. See Kevin Collier, 'Actors vs. AI: Strike brings focus to emerging use of advanced tech', July 14 2023, NBC News, available at: https://www.nbcnews.com/tech/tech-news/hollywood-actor-sag-aftra-ai-artificial-intelligence-strike-rcna94191.

[22] US District Court for the Northern District of California, decision of 30 October 2023, Case No. 23-cv-00201-WHO, Sarah Andersen et al. v. Stability AI.

[23] European Commission, proposal for a regulation of the European Parliament and of the Council laying down harmonised rules on Artificial Intelligence (Artificial Intelligence Act) and amending certain Union legislative Acts, 21 April 2021, COM(2021) 206. More in general, the proposal embraces transparency as a core benchmark in the governance of AI following the work of the Council of Europe on AI, namely the Resolution of 22 October 2020 No. 2344, 'The brain-computer interface: new rights or new threats to fundamental freedoms?', para. 6.7; Resolution of 22 October 2020, No. 2345, 'Artificial intelligence and labour markets: friends or foe?', paras. 7, 8; Resolution of 22 October 2020, 'Need for democratic governance of artificial intelligence', No. 2341, paras. 5, 14, 16.

and the judiciary have started to decide on the copyrightability of AI-generated outputs.[24] Despite the different constitutional frameworks and copyright legislations in force on both sides of the Atlantic, there is a common trend to reject algorithmic authorship based on the historical anthropocentric approach to copyright law.[25] It is very likely that many more cases will be brought to courts in the near future. The most problematic litigations will presumably concern "mixed" works resulting from the combination of human and artificial contributions, being necessary to determine the exact threshold above which the algorithmic participation in the creative process outshines the human involvement in a way that it cannot longer trigger copyright protection.[26] In the US, there might be some help from the guidance that the Department of Commerce will develop for detecting and watermarking (purely) AI-generated content to implement the recent Executive Order issued by President Biden aimed at ensuring 'Safe, Secure, and Trustworthy Artificial Intelligence'.[27] The guidelines will contribute to exclude the risk of false human attribution of content entirely produced by AI. However, it is reasonable to assume that the remaining works still necessitate an assessment of the human participation in the creative process.[28]

This article is focused on the input side of the challenges raised by generative AI. Drawing on a previous paper and in line with some recent proposals advanced in IP literature (see below), it suggests exploring the idea of introducing a statutory license for machine learning purposes as a compromise solution to ensure an attractive environment for artificial intelligence without marginalizing the role played by human authors.[29] This remuneration proposal is rooted on a fundamental rights analysis that balances the competing interests at stake.

2. The belated copyright provision in the artificial intelligence act and the unfitted current text and data mining provisions for generative AI

In its original version, the AIA did not address copyright aspects. It aimed at striking a balance between enhancing innovation while securing fundamental rights through the adoption of a risk-based approach that was (quite surprisingly) totally agnostic to intellectual property rights. However, the outlined tensions between providers of generative AI and copyright holders led the European Parliament to include some limited considerations with regard to copyright aspects of machine learning.

Firstly, the amendment 399 to Art. 28 b) offers a notion of generative AI before national legislators engage in their own defining attempts. Indeed, the endemic cross-border applications of this technology make a fragmented approach highly undesirable. The European Parliament proposed to define generative AI as the service provided through 'foundation models used in AI systems specifically intended to generate, with varying levels of autonomy, content such as complex text, images, audio, or video'. The illustrative list of intellectual works is to be welcomed because it enhances the adaptability of the provision to the growing production capacities of generative AI which in the future may cover any kind of creative segment.

Secondly, para. 4, lett. C) of the mentioned amendment imposes on providers of foundation models the obligation to train, design and develop the model in compliance with Union and national legislation on copyright prior to making their service available on the market. To this purpose, providers should publish a sufficiently detailed summary of the use of training data protected under copyright law.

[24] In 2019, the U.S. Copyright Office (USCO) denied copyright protection for a painting titled "A Recent Entrance to Paradise" allegedly realized by the AI system named "Creativity Machine" because the work lacked human authorship. The decision was confirmed by the Review Board of the Copyright Office in February 2022 as well as in the recent decision by the U.S. District Court of Columbia of 18 August 2023, No 22-1564, specifying that 'human authorship is a bedrock requirement of copyright'. On 21 February 2023, the USCO reviewed the registration of the comic book "Zarya of the Dawn" (Registration No. VAu001480196) excluding copyright protection for the images produced by the AI system Midjourney on the grounds that the changes by the alleged author were "too minor and imperceptible to supply the necessary creativity for copyright protection". The USCO, in the Copyright Registration Guidance: Works Containing Material Generated by Artificial Intelligence, 16 March 2023, No. 88 FR 16190, confirmed the restrictive approach against the registration of works lacking any (substantial) human contribution. Moreover, the Italian Supreme Court, in the decision no. 1107 of 16 January 2023, acknowledged copyright protection for a digital flower created with the aid of a software because the human contribution of the author was still identifiable. As a software-implemented creation, it was not in the public domain and the company willing to exploit the work had to clear the right to reproduction. In China, the Beijing Internet Court denied copyright protection for an AI-generated work because of the lack of human involvement in the creative process. However, in another judgment of 24 December 2019, the Nanshan District Court of Shenzhen awarded copyright to an AI-generated text since it complied with the formal requirement of written work. For a comparison of the two decisions see, Yong Wan and Hongxuyang Lu, 'Copyright protection for AI-generated outputs: the experience from China' (2021) 42 Computer Law & Security Review.

[25] See on this issue Daniel Gervais, 'The Machine as Author', Iowa Law Review, 2020, Vol. 105, p. 2053; Jane C. Ginsburg, 'People Not Machines: Authorship and What It Means in the Berne Convention' (2018) 49 International Review of Intellectual Property and Competition Law, 131; Jane C. Ginsburg and Luke Ali Budiardjo, 'Authors and Machines' (2019) 34(2) Berkeley Technology Law Journal, 343.

[26] See, for instance, the decision by the Review Board of the USCO on 5 September 2023 which confirmed the refusal to register the work generated by Midjourney "Théâtre D'opéra Spatial" (SR No. 1-11743923581), a piece that garnered national attention for winning the 2022 Colorado State Fair's annual fine art competition. Despite the applicant argued to have input at least 624 prompts, the office rejected the request for reconsideration because it was not possible to separate the human contribution from the algorithmic one. It can be deducted that only giving instructions to the machine via prompts is not considered as being sufficient to establish substantial human creativity. The decision is available at the following link: copyright.gov/rulings-filings/review-board/docs/Theatre-Dopera-Spatial.pdf.

[27] The White House, Fact Sheet: President Biden Issues Executive Order on Safe, Secure, and Trustworthy Artificial Intelligence, 30 October 2023, available at the following link: https://www.whitehouse.gov/briefing-room/statements-releases/2023/10/30/fact-sheet-president-biden-issues-executive-order-on-safe-secure-and-trustworthy-artificial-intelligence/

[28] See in particular Kateryna Militsyna, 'Human Creative Contribution to AI-Based Output – One Just Can('t) Get Enough' (2023) 72(10) GRUR International 939-949, who proposes an interesting five part-test to distinguish between sufficient and insufficient human creative participation to benefit from copyright protection (test that could probably be simplified as the third factor, intended by the Author as the evaluation of free and creative choices, seems to overlap with the fifth factor which is devoted to the originality requirement). It should be recalled that the CJEU has several times clarified that originality in copyright law occurs when the author expresses free and creative choices able to reflect its personality). For further reflections on the suitability of AI-generated works to meet the originality requirement see Vincenzo Iaia, 'To Be, or Not to Be…Original Under Copyright Law, That Is (One of) the Main Questions Concerning AI-Produced Works' (2022) 71(9) GRUR International 793-812.

[29] Christophe Geiger, 'When the Robots (try to) Take Over: Of Artificial Intelligence, Authors, Creativity and Copyright Protection', in Florent Thouvenin, Alexander Peukert, Thomas Jäger and Christophe Geiger (eds.), Innovation - Creation - Markets, Festschrift Reto M. Hilty, (Springer, forthcoming, 2024). See already, but more broadly, Christophe Geiger, 'Freedom of Artistic Creativity and Copyright Law: A Compatible Combination?', (2018) 8(3) U.C. Irvine L. Rev. 413, advancing a proposal for a limitation-based statutory remuneration for commercial creative re-use of copyright protected works grounded in fundamental rights and in particular in freedom of artistic creativity and the right of authors to the protection of their material interests.

The transparency provision seems to require providers of foundation models to disclose a comprehensive listing of the copyrighted content used for training their algorithm(s), accompanied by precise identification of rightsholders. In this context, it needs to be recalled that training the AI algorithm is done by using text and data mining technologies ('TDM'), the use of which have been regulated by the EU in 2019 with the enactment of the CDSM directive.[30] If the objective of the intervention was not to regulate Generative AI (the issue was on the contrary totally overlooked[31]), the European legislator was aware that automated computational analysis of information could involve works protected by copyright or by *sui generis* database right, thus potentially infringing upon the exclusive reproduction right.

Apart from the definition of TDM under Art. 2.2, intended as "any automated analytical technique aimed at analysing text and data in digital form in order to generate information which includes but is not limited to patterns, trends and correlations", the directive implements two exceptions and limitations to copyright law in Arts. 3 and 4 of the CDSM directive.[32] Both mandate Member States to introduce a TDM exception or limitation which curtails the exclusive power arising from copyright or *sui generis* right. On the one hand, Art. 3 entitles research organizations and cultural heritage institutions to carry out text and data mining of all the works to which they have lawful access for the purposes of scientific research. On the other hand, Art. 4 is not limited with regard to specific institutions or research objectives as it legitimates any entity to perform TDM activities for commercial purposes under the condition of lawful access to such works. However, there is an important limit which excludes from the scope of the exception/limitation those works that have been expressly reserved by their rightsholders in an appropriate manner, such as (still unharmonized) machine-readable means (Art. 4.3), which as many scholars have underlined will likely make this exception unworkable,[33] in particular in the AI context where the opt-out might lead to incomplete training data sets and legal uncertainties around AI-generated works.[34]

It can be presumed that the transparency rules set forth by the AIA have been introduced to allow rightsholders to more effectively exercise the opt-out right from the text and data mining exception established by Art. 4.3 of the directive and control its enforcement. Indeed, without any reporting obligation, authors and rightsholders have no means to trace the supply chain behind AI-generated outputs. The introduction could also be interpreted as a first step for the EU legislator to establish a future obligation to obtain a license for the ML uses in question, should it be considered that these uses fall within the exclusive right. This seems to be the purpose of several lawsuits engaged against AI system producers in the US, claiming that these uses do not qualify as fair use under the US Copyright law. In the US, it is generally considered[35] that TDM is s a fair use based on the application by analogy of *Authors Guild v. Google*[36] and *Authors Guild v. HathiTrust.*[37] However, it is debated whether Generative AI training cuts against the fourth factor (the effect of the use upon the potential market[38]) in light of the direct competition that AI-generated works can have with those created by physical persons which may lead to chilling effects on human creativity.[39] The various lawsuits taken in the U.S. will soon bring first elements on the legality of harvesting in-copyright works for ML purposes. Some great pressure from rightsholders can surely be expected in this regard on this side of the Atlantic as well.

A good example of a maximalist approach is reflected in the recent draft bill proposal introduced in France on September 12, 2023, which proposes to submit the machine learning process to the exclusive control of the rightsholders whose works are used and that the authorship of the works should be attributed to the authors of the works used in the machine learning process.[40] Further, it obliges the creator to label the generated output "AI generated work" and to list the names of all

[30] Directive (EU) 2019/790 of the European Parliament and of the Council of 17 April 2019 on copyright and related rights in the Digital Single Market and amending Directives 96/9/EC and 2001/29/EC, OJ L 130.

[31] See Christophe Geiger, 'The missing goal-scorers in the artificial intelligence team: of big data, the fundamental right to research and the failed text and data mining limitations in the CDSM Directive', supra.

[32] For an in-depth analysis of the text and data mining exception see especially Martin Senftleben, 'Study on EU copyright and related rights and access to and reuse of data' (European Commission, 2022); Christophe Geiger, Giancarlo Frosio & Oleksandr Bulayenko, 'Text and Data Mining: Articles 3 and 4 of the Directive 2019/790/EU', in: Conception Saiz Garcia & Raquel Evangelio Llorca (eds.), Propiedad intelectual y mercado único digital europeo, Tirant lo Blanch, 2019, 27.

[33] See in this regard Christophe Geiger, Giancarlo Frosio & Oleksandr Bulayenko, 'The Exception for Text and Data Mining (TDM) in the Proposed Directive on Copyright in the Digital Single Market - Legal Aspects, Study for the Directorate-General for Internal Policies of the Union', Department of Citizens' Rights and Constitutional Affairs, European Parliament, February 2018, and from the same authors 'Crafting a Text and Data Mining Exception for Machine Learning and Big Data in the Digital Single Market', in: Xavier Seuba, Christophe Geiger and Julien Pénin (eds.), 'Intellectual Property and Digital Trade in the Age of Artificial Intelligence and Big Data', Global Perspectives and Challenges for the Intellectual Property System, No. 5, Geneva/Strasbourg, 2018, p. 95. For a critical evaluation of the Directive proposal, see also Christophe Geiger, Giancarlo Frosio and Oleksandr Bulayenko, 'Text and Data Mining in the Proposed Copyright Reform: Making the EU Ready for an Age of Big Data?' (2018) 49(7) International Review of Intellectual Property and Competition Law 814, and from the same authors: 'The EU Commission's Proposal to Reform Copyright Limitations: A Good but Far Too Timid Step in the Right Direction' (2018) 40 European Intellectual Property Review 4.

[34] For instance, Sacem (Société des auteurs, compositeurs et éditeurs de musique), the major collective management organization for music in France, announced that it will exercise the opt-out on behalf of all its 200.000 members, making it probable that other collective management organizations in France and other European countries will follow the same path. This reaction would be particularly pernicious for the development of AI-systems considering that a substantial part of works would be left out from the training pool for ML purposes. See the official communication by Sacem, 'Pour une intelligence artificielle vertueuse, transparente, et equitable: la Sacem exerce son droit d'opt-out', 12 October 2023, available at the following link: https://createurs-editeurs.sacem.fr/actualites-agenda/actualites/la-sacem-et-vous/pour-une-intelligence-artificielle-vertueuse-transparente-et-equitable-la-sacem-exerce-son-droit.

[35] Carys J. Craig, 'The AI-Copyright Challenge: Tech-Neutrality, Authorship, and the Public Interest', in: Ryan Abbott (ed.) Research Handbook on Intellectual Property and Artificial Intelligence (Edward Elgar, 2022) 152; Pamela Samuelson, 'Text and Data Mining of In-Copyright Works: Is It Legal?' (2021) 64(1) Communications of the ACM 20-22; Mark A. Lemley and Bryan Casey, 'Fair Learning' (2021) 99(4) Texas Law Review, 743-785; Michael Carroll, '*Copyright and the Progress of Science: Why Text and Data Mining Is Lawful',* 53 U. C. Davis L. Rev. 893 (2019)

[36] U.S. Court of Appeals for the Second Circuit, Authors Guild, Inc. v. Google, decision of 16 October 2015, No. 804 F.3d 202.

[37] U.S. Court of Appeals for the Second Circuit, Authors Guild, Inc. v. HathiTrust, decision of 10 June 2014, No. 755 F.3d 87.

[38] The four factors are listed in Section 107 of the 1976 U.S. Copyright Act.

[39] Even if TDM could qualify as a fair use in the past, it remains uncertain if the use of copyrighted works via machine learning in generative AI processes will be allowed under the fair use doctrine (see also more detailed on the issue of generative AI and fair use in the US: Matthew Sag, 'Copyright Safety for Generative AI' (2023) 61(2) Houston Law Review; forthcoming).

[40] French National Assembly, Draft Law of the 12 September 2023, No. 1630, available at the following link: https://www.assemblee-nationale.fr/dyn/16/textes/l16b1630_proposition-loi.pdf.

authors whose works have been used in the training process. This would require crediting thousands – if not millions – of authors since it is now well established that Generative AI is trained on gigantic corpora of data. From a policy perspective, such an overreaching solution, no matter how well intended, would be very detrimental to the development of AI systems and result in making the jurisdiction adopting it very unattractive in these innovative sectors. The atavistic fear of disruptive technologies resulting from a partial comprehension of their opportunities risks nourishing the appetite for regulatory measures that discourage the development in strategic industries and new technologies. In sum, even we certainly agree that a "humanist principle"[41] should guide the governance of AI, the French proposal appears quite myopic inasmuch it does not take into account the inevitable existence of AI in our daily lives and its power to augment human creativity by the assistance of AI tools to uncharted levels.

Furthermore, it has been rightly stressed that the fulfillment of the transparency obligation with regard to the works used appears quite unfeasible because of the low, still inhomogeneous, threshold of originality, the fragmentation of copyright across various jurisdictions and its multiple ownerships, the absence of a mandatory registration process and the general inadequate state of ownership metadata.[42] Also, the technical feasibility needs to be confirmed as algorithms can be trained from an immense variety of sources and it might not always be easy to determine precisely which sources have been used.[43] The issue here is to find an efficient solution to the trade-off between an appropriate level of transparency and an appropriate level of feasibility.[44]

It becomes crucial to elucidate the specific content of this proposals in the AI act during trilogue negotiations. The rationale behind the latest amendment to the AIA is quite intuitive: ensuring collaboration between providers of generative AI services and copyright holders as regards to this new form of exploitation of creative works. The great divergence of interests and the high transaction costs of a potential licensing solution makes it unlikely, however, that agreed solutions can be elaborated without future legislative intervention.[45] It is also probably not desirable that this crucial question for the future of creativity in the digital environment is left solely to the self-regulation of the various market players. In addition, although some commentators call on creators to have a very proactive agenda to protect their rights,[46] in practice creators are often recalcitrant to engage in costly and time-consuming litigations, not having the means to monitor and enforce their rights.

Moreover, the amendment introducing transparency with regard to the works use to train the AI seems to provide an effective enforcement mechanism to the opt-out right set forth by Art. 4.3 of the CDSM Directive. In the absence of a report containing all the copyrighted works mined and extracted for machine learning ('ML') purposes, it would be nearly impossible for rightsholders to discover that their work has been injected into the software, except for blatant cases where the initial work is recognizable in the AI output or when there are other clear indications, such as the image produced by Stability AI that showed two football players with a watermark very similar to that of Getty Images.

However, the provision may produce an unintended – or at least undesirable – consequence: a sharp cut in the datasets available for the algorithmic training resulting from a massive exercise of the opt-out right. This would in turn throw sand in the gears of AI-systems[47] by affecting the quality of the AI-produced outputs according to the old adage in information systems "garbage in, garbage out".[48] The narrative on biases in AI is rich with examples of the nexus between flawed inputs and flawed outputs, such as the stereotyped representation of female nurse vs. male doctor.[49]

It is a delicate balancing exercise because the introduction of excessive (administrative and/or financial) burdens on AI-providers may limit the input datasets with consequences on the advancement of AI systems. Indeed, the value of Generative AI services to ensure support for creative activities shall not be underestimated. Some artists are already using this technology as a vehicle to discover new esthetic and artistic languages.[50] Also, it should not be forgotten that Generative AI can also be used for scientific purposes, which might call for differentiated approaches with regard to the purpose of the ML uses in question[51] as the fundamental right to research calls for a privileged treatment of research over copyright claims.[52] The main challenge is to

[41] The introductory motives of the proposal put such a principle forward to justify the legislative action. It reads : "Face à l'IA, écosystème qui avance à pas de géant, le législateur doit protéger impérativement les auteurs et artistes de la création et de l'interprétation selon un 'principe humaniste'" (emphasis added).

[42] João Pedro Quintais 'Generative AI, Copyright and the AI Act' Kluwer Copyright Blog, 9 May 2023, available at: https://copyrightblog.kluweriplaw.com/2023/05/09/generative-ai-copyright-and-the-ai-act/

[43] See Jan Bernd Nordemann, Jonathan Pukas, 'Copyright exceptions for AI training data – will there be an international level playing field?' (2022) 17(12) Journal of Intellectual Property Law & Practice 973-974, who points out that 'AI works as a black box with the underlying patterns not being output to the AI developers in a comprehensible format'.

[44] In this respect, maybe a "best effort" standard to provide for transparency on the training data in the spirit of Art. 17 of the CDSM directive rather than a strict obligation would be a more workable first step towards a satisfactory balance of the different needs.

[45] There are some exceptional cases, such as Adobe Firefly AI, where the company has recently established a compensation structure for contributors whose images have been used for training the algorithm. However, the early roll out of the service impedes to assess how virtuous this practice is for content creators. For more details see Johan Moreno, 'Adobe Breaks Firefly AI Out of Beta, Rolls Out Generative AI Capabilities Across Apps' Forbes, 13 September 2023, available at the following link: https://www.forbes.com/sites/johanmoreno/2023/09/13/adobe-breaks-firefly-ai-out-of-beta-rolls-out-generative-ai-capabilities-across-apps/?sh=7c4886dd7d05.

[46] See Gil Appel, Juliana Neelbauer and Davide Schweidel, 'Generative AI Has an Intellectual Property Problem' Harvard Business Review, 7 April 2023, available at https://hbr.org/2023/04/generative-ai-has-an-intellectual-property-problem. The Authors invite creators to look if their works are compiled in large-scale data lakes and to monitor digital and social channels for the circulation of potential unauthorized derivative works.

[47] See in these eloquent terms Gustavo Ghidini, 'Proprietà intellettuale e innovazione digitale. Dalla 'interferenza antitrust' a un nuovo paradigma?', in Giurisprudenza Commerciale, III, 2023, p. 369.

[48] Further on this principle Gürol Canbek, 'Gaining insights in datasets in the shade of "garbage in, garbage out" rationale: Feature space distribution fitting' (2022) 12(3) Data Mining and Knowledge Discovery 1-18.

[49] Kaśka Porayska-Pomsta, Wayne Holmes, Selena Nemorin, 'The ethics of AI in education', in Antonija Mitrovic, Benedict du Boulay, Kalina Yacef (eds.) Handbook of Artificial Intelligence in Education (Edward Elgar, 2023) 575.

[50] See, for instance, the artworks by Davide Quayola that materialize from the use of various technologies for purposes other than those for which they were conceived: https://quayola.com/exhibitions/. See also Refik Anadol's work "Unsupervised", exposed at MoMA in New York until 29 October 2023: https://www.moma.org/calendar/exhibitions/5535. In this perspective, machines are source of inspiration for human beings which in turn can act as a catalyst for new artistic expressions.

[51] See James Love, 'We Need Smart Intellectual Property Laws for Artificial Intelligence' Scientific American, 7 August 2023, available at: https://www.scientificamerican.com/article/we-need-smart-intellectual-property-laws-for-artificial-intelligence/.

[52] Christophe Geiger and Bernd Justin Jütte, "Conceptualizing a 'Right to Research' and its Implications for Copyright Law, An International and European Perspective" (2023) 38(1) American University International Law Review 1; with Sean Flynn, Christophe Geiger, Joao Pedro Quintais et al., 'Implementing User Rights for Research in the Field of Artificial Intelligence: A Call for International Action', European Intellectual Property Review 2020, Vol. 42, Issue 7, 393

lay down a legal framework in which AI-based tools remain instrumental to human creativity rather than a stronger substitute. In addition, there remain some doubts as to how AI companies will operationalize the reporting obligation under the belated copyright provision.

3. Statutory licenses as a compromise solution to consider the interest of AI developers as well as the remuneration interest of creators

Various solutions are emerging with the common aim of reconciling the need of AI developers for massive and accurate data with the demand of copyright holders for equitable remuneration. Some authors advocate for the establishment of data-sharing agreements with data providers or for royalty-based compensation models in direct contact with content creators,[53] perhaps accompanied by tools able to verify the training through a sort of reverse engineering mechanism.[54]

However, the gigantic transaction costs of one-to-one negotiations and the risks of having incomplete datasets inhibiting the developments of performing AI systems leads to envisaging other solutions that reconcile more efficiently the interest of AI developers and the remuneration interest of creators, such as the introduction of a remunerated copyright limitation that imposes to the benefit of human creators a general payment obligation on providers of generative AI systems for the use of copyrighted works for ML purposes in the context of Generative AI systems.[55] From this perspective, it can be argued that the introduction of a limitation-based remuneration right will preserve the anthropocentric approach to copyright law, preventing an upside-down scenario where human creativity becomes instrumental to artificial intelligence.[56] Such a limitation-based remuneration right would replace the unworkable opt-out option foreseen in Art. 4 of the CDSM directive for commercial TDM activities outside of the scope of Art. 3.[57]

It is worth recalling that statutory licenses are limitations of the exclusive power arising from copyright protection that are justified by the objective of securing derivative creativity in specific sectors that otherwise are likely to be stagnant.[58] The US copyright system has developed this exceptional "permitted-but-paid"[59] regime which curtails the exclusive rights of copyright owners for specific aims, namely preventing monopoly in the music sector and reducing transaction costs for the licensing of sound recordings and television programs.[60]

In the EU, Art. 5.2 b of the InfoSoc Directive[61] introduced a remunerated private copying exception or limitation, a similar mechanism that Member States could implement to offset the damage that rightsholders suffer for the private reproduction of their works. As in the case of Generative AI, content creators were not able to monitor and enforce their rights while their works were used without remuneration. This scenario brought the German legislator to address the problem of the fair remuneration of copyright holders in uncontrollable environments by establishing the first levy system in the 1965 Copyright Act.[62] The payment is integrated into a levy that is imposed on the cost of a physical medium or a device which allows the user to duplicate copyright protected works. Then, the collected levies are distributed among content creators and rightsholders based on different criteria across the Member States that have implemented Art. 5.2 b.[63] Applied to ML, the remuneration would result from the machine learning uses of copyright protected works by an algorithm for commercial Generative AI purposes.

Interestingly, the Italian legal system also offers an example of such "permitted-but paid" use with regards to engineering projects that provide original solutions to technical problems. Those works are protected by a 20-years neighboring right lasting from the deposit of the work at the Office of the President of the Council. In particular, Art. 99 of the Italian Copyright Law[64] entitles the author of such works to be fairly compensated for the unauthorised for-profit implementation of the technical project. The justification behind the limitation of the author's power lies in the interest in fostering technical progress in the engineering sector[65] as well as in that of preventing that safer techniques for the construction of bridges, buildings and other structures remain appanage of their creators or rightsholders, typically the construction companies.[66] Hence, the Italian legislator decided to remove obstacles that could hinder innovation and safety in this specific field.

However, more compelling justifications for introducing a statutory remuneration right for commercial ML purposes in place of the opt-out right under Art. 4.3 of the CDSM Directive can be found in the fundamental rights framework. Indeed, the question of Generative AI vs.

[53] Nicola Lucchi, 'ChatGPT: A Case Study on Copyright Challenges for Generative Artificial Intelligence Systems' (2023) European Journal of Risk Regulation 1-23.

[54] Alain Strowel, 'ChatGPT and Generative AI Tools: Theft of Intellectual Labor?' (2023) 54(4) International Review of Intellectual Property and Competition Law 494.

[55] Christophe Geiger, 'When the Robots (try to) Take Over: Of Artificial Intelligence, Authors, Creativity and Copyright Protection', supra and in the context of the use of journalistic content Christophe Geiger and Bernd Justin Jütte, 'Designing Digital Constitutionalism: Copyright Exceptions and Limitations as a Regulatory Framework for Media Freedom and the Right to Information Online', in: Martin Senftleben et al. (eds) "Cambridge Handbook of Media Law and Policy in Europe" (Cambridge University Press, forthcoming, 2024), available on ssrn: https://ssrn.com/abstract=4548510; Gustavo Ghidini, 'Proprietà intellettuale e innovazione digitale. Dalla 'interferenza antitrust' a un nuovo paradigma?', supra, 367-375 Similar proposals have been advanced recently on the input or on the output of AI systems by Giancarlo Frosio, 'Should We Ban Generative AI, Incentivise It or Make it a Medium for Inclusive Creativity', in Enrico Bonadio, Caterina Sganga (eds.), *A Research Agenda for EU Copyright Law* (Edward Elgar, forthcoming, 2024); for similar considerations linked to the AI-generated output see Martin Senftleben, 'Generative AI and Author Remuneration', 14 June 2023, available at SSRN at the following link: https://ssrn.com/abstract=4478370.

[56] Christophe Geiger, 'When the Robots (try to) Take Over: Of Artificial Intelligence, Authors, Creativity and Copyright Protection', cit.

[57] The introduction of such a statutory remuneration should therefore go hand in hand with a reform of article 3 and 4 CDSM and could be also the opportunity to remedy some of the deficiencies of these provisions with regard as legitimate uses with strong freedom of expression/ right to information purposes such as journalistic uses of TDM (See in the sense in particular Christophe Geiger, 'Designing Digital Constitutionalism: Copyright Exceptions and Limitations as a Regulatory Framework for Media Freedom and the Right to Information Online', in Martin Senftleben et al. (eds.), Cambridge Handbook of Media Law and Policy in Europe (Cambridge University Press, forthcoming).

[58] For deeper considerations on this tool see Christophe Geiger, 'Statutory Licenses as Enabler of Creative Uses' in Kung-Chung Liu, Reto Hilty (eds.), Remuneration of Copyright Owners (Springer, 2017), 305-327.

[59] This terminology in the context of copyright limitations has been coined by Jane Carol Ginsburg, 'Fair Use for Free, or Permitted-but-Paid' (2014) 29 Berkeley Technology Law Journal 1446.

[60] See especially Sections 111, 112, 114, 115, 116, 118, 119, and 122 of the 1976 U.S. Copyright Act.

[61] Directive 2001/29/EC of the European Parliament and of the Council of 22 May 2001 on the harmonisation of certain aspects of copyright and related rights in the information society, L 107.

[62] See the extended research by Martin Kretschmer, 'Private Copying and Fair Compensation: An empirical study of copyright levies in Europe' 2011 The Intellectual Property Office Research Paper No. 2011/9.

[63] For a comparative overview of the permitted-but-paid uses established in European countries see especially Jane C. Ginsburg, 'Fair Use for Free, or Permitted-but-Paid', cit. 1414-1423.

[64] Law of 22 April 1941, no. 633.

[65] Antonino Cataudella, 'Rilettura delle norme a tutela degli autori dei progetti di ingegneria' (1982) 1(4) Rivista del Diritto Commerciale e del diritto generale delle Obbligazioni.

[66] Gustavo Ghidini, 'Proprietà intellettuale e innovazione digitale. Dalla 'interferenza antitrust' a un nuovo paradigma?', cit., 373.

authors' remuneration opens the way to statutory licenses as they provide a compelling balance of the different fundamental rights involved.

The justification of this exceptional regulatory instrument is rooted on Arts. 11 and 13 EUCF, 19 UDHR, 27.1 UDHR, 15.1 a and b ICESCR, considering that Generative AI training is essential for human beings to explore new avenues of artistic expression that are still unknown. Indeed, some outputs produced by generative AI have the potential to be used by human authors in new forms of art and cultural expressions that benefit the society at large. Access to comprehensive data, even if protected by copyright, is therefore a precondition for the correct operation and advancement of AI systems. Without a general limitation covering any copyrighted content, the productive capacity of Generative AI-based tools – and their power to assist human beings reaching new artistic and scientific results – risks being left partially unrealized. Indeed, a limited training dataset deriving from technical or legal (copyright) constraints cannot but marginalizes the heterogeneity of AI-generated outputs, thus increasing their homologation rate and the likelihood of biases. It should be made clear that freedom of artistic expression concerns exclusively human beings considering that, at least under the current state of the law, AI does not enjoy the mentioned constitutional right, in particular no right to free expression.[67] This implies that the interest in the flourishing of the Generative AI industry should remain instrumental to the objective of increasing human artistic freedom of expression.

The functioning of statutory licenses allows maximization of the copyright content exploitable for ML purposes while taking into account the interests of the authors to be remunerated for the commercial use of their intellectual efforts, as protected by Arts. 17.2 EUCF, 27.2 UDHR, and 15.1 c ICESCR. AI developers should thus share their revenues with the authors whose works are used to train the algorithms. A right to a fair remuneration granted by transparency obligations encourages human beings to still produce new works (eventually with the expanded creative ability that generative AI models can provide) while securing that the use of their work by AI systems generates a fair return. The human creator remains in this way at the center of the copyright system.

As emphasized previously, 'a conceptualization of copyright within the constitutional right to science and culture, to freedom of expression and artistic creativity and even within a socially rooted property clause […] should lead to the recognition of an overarching and unwritten right for creators to be remunerated for the commercial exploitation of their work. This right should be understood as a fundamental and binding principle of copyright law deriving from fundamental rights and from copyright's social function and would thus have to be recognized by legislators through the implementation of appropriate mandatory copyright contract rules or statutory remunerations rights; it could also be used by the judiciary as an interpretation tool of existing rules to redress unfair remuneration situations for creators, or, in their absence, even beyond as a general principle of law'.[68]

The advocated statutory license is the tangible embodiment of the suggested approach to address copyright issues related to Generative AI in a fundamental rights-compliant manner. Indeed, it enhances a complementary dialog between the imperative of access to in-copyright works (which in this case is a technical necessity for the development of AI systems) and the protection of the moral and material interests of creators.[69] Doing so, it can draw but also help defining on the theoretical framework of "digital constitutionalism", an emerging theory aimed at operationalizing traditional constitutional values in the contexts of digital technologies and online environments.[70] Since the advent of the Internet, these were the spaces showing more permeability to the powers of private actors due to weaknesses of constitutional protection, jeopardizing the fundamental rights of individuals.[71] The theory of digital constitutionalism builds exactly on fundamental rights as a backbone of any modern constitutional democracy to address the loopholes and the imbalances originating in the digital ecosystems. From the perspective of digital constitutionalism, which is concerned with the effective realization of fundamental rights in the digital environment, the risk that AI developers piggyback on creators without sharing the revenues linked to the prosperous Generative AI market needs to be addressed from a fundamental right angle. The introduction of a remunerated copyright limitation for Generative AI commercial purposes represents a compelling solution to meet the endangered remuneration right of creators without disproportionately sacrificing the interest of AI developers to offer increasingly high-performing services able to foster human creativity.

This policy option would also most likely be in compliance with the three-step test established by Art. 9 of the Berne Convention and Art. 13 TRIPs. This international standard allows to evaluate the permissibility of the introduction of a limitation and exception with regard to the following conditions: i) its scope should be confined to certain special cases; ii) it does not conflict with a normal exploitation of the work; iii) it does not unreasonably prejudice the legitimate interests of the author (Berne) / rightsholder (TRIPS). It is not possible here to provide for a detailed analysis, however a strong line of scholarship has argued that the fair remuneration component helps to overcome the problem of the hurdle of the second and third step, in particular when the use has a strong fundamental rights justification.[72]

Also, from a systematic point of view, an easier access of information

[67] Cass. R. Sunstein 'Artificial Intelligence and the First Amendment', 28 April 2023, available at SSRN: https://ssrn.com/abstract=4431251. It is hoped that legislators worldwide will not be tempted by the sirens of technocratic investment protection that for several years have also informed regulatory interventions in the field of intellectual property. As noted by Christophe Geiger, 'Intellectual Property and Investment Protection: A Misleading Equation', in Viktor Fischer, George Nolte, Martin Senftleben, Louisa Specht-Riemenschneider (eds.), Gestaltung der Informationsrechtsordnung – Festschrift fur Thomas Dreier zum 65 (Beck, 2022) 7, 'creativity rather investment should be triggering factor for protection'.

[68] See Christophe Geiger, 'Building an ethical framework for intellectual property in the EU: time to revise the Charter of Fundamental Rights', in in Gustavo Ghidini and Valeria Falce (eds.), Reforming Intellectual Property Law (Edward Elgar, 2022), 90-91.

[69] In this sense, Christophe Geiger, 'Copyright as an access right: Securing cultural participation through the protection of creators' interests' (2015) Max Planck Institute for Innovation & Competition Research Paper No. 15-07.

[70] Further on this theory in the IP context, see Christophe Geiger and Bernd Justin Jutte, 'Designing Digital Constitutionalism: Copyright Exceptions and Limitations as a Regulatory Framework for Media Freedom and the Right to Information Online', supra, and the referred bibliography; Christophe Geiger & Bernd Justin Jütte, 'Digital Constitutionalism and Copyright Reform: Securing Access to through Fundamental Rights in the Online World', The Digital Constitutionalist, 24.01.2022, available at:http://digi-con.org/digital-constitutionalism-and-copyright-reform-securing-access-through-fundamentalrights- in-the-online-world/

[71] Giovanni De Gregorio, Digital Constitutionalism in Europe, Reframing Rights and Powers in the Algorithmic Society (Cambridge University Press, 2022), 88-89. See also more generally Oreste Pollicino, Judicial Protection of Fundamental Rights on the Internet, A Road towards Digital Constitutionalism? (Bloomsbury Publishing, 2021).

[72] See for further reference Christophe Geiger, Daniel Gervais, and Martin Senftleben, 'The Three-Step Test Revisited: How to Use the Test's Flexibility in National Copyright Law', American University International Law Review 2014, Vol. 29, Issue 3, 581. See also in particular the "Declaration on a balanced interpretation of the Three-Step Test in Copyright Law", IIC 2008, 707

embedded in copyright contents well aligns with the European strategy to ensure the fifth freedom of circulation of non-personal data.[73]

The concrete operation of the limitation-based remuneration for ML for commercial purposes could take two forms, depending on the preference for a private or public integration. In the first place, the quantification of remuneration rates can be left to negotiation. Collective management organizations can be tasked with the administration and redistribution of the statutory remuneration, and they might be able to secure better conditions because of their stronger bargaining position compared to single creators or rightsholders. On the other side, AI companies are expected to enter into negotiations in good faith to secure creators a share of the income linked to the growing market of generative AI services. As a backstop to this mechanism, the European Union could appoint a specialized authority on copyright/digital services issues to break the deadlock for the determination of the fair compensation that rightsholders are entitled to pretend for TDM activities.[74] In the absence of a needed regulatory body at EU level, national regulators could step in. For instance, in Italy, the Authority for Communications (AGCOM) could be appointed for this task since, with the implementation of the CDSM directive, it has been assigned a key role for setting the disputes between rightsholders and online content sharing service providers, also regarding the equitable compensation for the online exploitation of in-copyright works and press publications.[75]

An alternative path is the statutory definition of the parameters for determining the remuneration rates. In this context, the legislator or public authorities appointed by the legislator[76] should set the criteria to quantify an equitable remuneration for Generative AI training. Extending the principles of appropriate and proportionate remuneration set forth by Art. 18 of the CDSM Directive, it is reasonable to calculate the amount due to rightsholders according to the actual or potential economic value of the work and all the other circumstances of the case, such as the market practices of the relevant sector.[77]

In order to reduce the risk that such a mechanism benefits primarily superstars from every creative sector, considering that a substantial portion of the requests may concern the imitation of the style of well-known celebrities, some empirical studies on the effective risk of discrimination between artists based on their popularity would be very useful to assess the need for recalibration by the system in place; redistribution systems can be expected to secure that remuneration flows to niche creators as well. However, in general terms, it is reasonable to believe that this policy option, if combined with mandatory collective rights management, could direct substantial revenue streams to human authors, thus improving their living and working conditions.[78]

4. Conclusions

The exponential development of generative AI comes with some legitimate concerns on the fair remuneration of authors when their protected works serve as input for commercial ML purposes.[79]

However, it should be remembered that copyright law has been coping with unprecedented forms of exploitation of intellectual works resulting from market practices or technological developments. When the existing legislation did not successfully address the new dysfunctionalities affecting a creative segment, authors experienced a sense of inequality and frustration as being deprived of the revenues arising from the new exploitation of their fruits of the labor. The EU has taken action to correct these market failures, for instance by introducing the rental right,[80] the private copying exception and the contractual protection of authors vs. publishers to ensure an appropriate share of the revenues obtained from information society service providers.[81] Against this background, European copyright law appears still in its nascent phase as regards to statutory licenses and their potential role in solving the conflict between exclusive rights and other concurring interests.[82]

The right to culture and science and freedom of artistic expression, as enshrined by European and international human rights law, can justify the introduction of a statutory license that will grant an appealing environment for AI creativity without jeopardizing the right of the human authors to be remunerated for the commercial use for their works. This compromise solution within the copyright system is the

[73] European Parliament News, 'Free flow of non-personal data: Parliament approves EU's fifth freedom', 4 September 2018, available at: https://www.eppgroup.eu/newsroom/news/free-flow-of-data-new-era-for-digital-economy-in-europe. One of the pillars on which the European data strategy is based is the proposal for a regulation on harmonized rules on fair access to and use of data (Data Act) presented by the Commission on 23 February 2022. The central goal of the proposal is to optimize the economic value of data by empowering a broader spectrum of stakeholders to manage their data without compromising the incentives to invest in data generation.

[74] Proposing such an EU institution, see Christophe Geiger and Natasha Mangal, 'Regulating Creativity Online: Proposal for an EU Copyright Institution', *GRUR International* 2022, Vol. 71, Issue 10, p. 933. See in this sense for commercial creative reuses, Christophe Geiger, 'Freedom of Artistic Creativity and Copyright Law: A Compatible Combination?', supra, stating that "the conclusion is that such a limitation-based system is a workable option, provided that a fair remuneration is secured to the original creators in case of commercial exploitation of the derivative work. This framework could be guaranteed through the implementation of an independent regulation authority or a Copyright council, which could check that access is granted at a fair price, taking into account the specificity of the work in question, such as the potential market for the derivative work, the fame of the original creator or work used, etc" (at p. 457).

[75] See the Italian Legislative Decree of 8 November 2021, No. 177. The Authority, with the resolution No. 3/23/CONS has issued the regulation setting the criteria for the determination of fair compensation for the online use of press publications. Many more resolutions are expected to come in the next months.

[76] These can be the same regulation authorities just mentioned, with the differences that they would be tasked by the legislator ex ante to set remuneration standards and not ex post in the absence of agreements of the parties.

[77] CDSM directive, Recital 73. On this point, see Eleonora Rosati, Copyright in the Digital Single Market: Article-by-Article Commentary to the Provisions of Directive 2019/790 (Oxford University Press, 2021) 360-367.

[78] More detailed see Christophe Geiger, 'When the Robots (try to) Take Over: Of Artificial Intelligence, Authors, Creativity and Copyright Protection', cit.

[79] A worldwide acknowledgment of these new challenges for the remuneration of creators comes from the action No. 11 of the Hiroshima Code of Conduct for Advanced AI Systems recently issued on 30 October 2023 by G7 leaders which expressly endorses AI service providers to implement appropriate safeguards to respect intellectual property, including copyright-protected content. The Code of Conduct (available at the following link: https://digital-strategy.ec.europa.eu/en/library/hiroshima-process-international-code-conduct-advanced-ai-systems) is allegedly based on the OECD AI Principles albeit they laconically acknowledge that the regulatory framework on intellectual property rights have relevance to AI.

[80] Directive 2006/115/EC of the European Parliament and of the Council of 12 December 2006 on rental right and lending right and on certain rights related to copyright in the field of intellectual property, OJ L 376.

[81] Art. 15.5, Directive (EU) 2019/790 of the European Parliament and of the Council of 17 April 2019 on copyright and related rights in the Digital Single Market and amending Directives 96/9/EC and 2001/29/EC, OJ L 130.

[82] See in this sense Christophe Geiger, 'Promoting Creativity through Copyright Limitations, Reflections on the Concept of Exclusivity in Copyright Law' (2010) 12(3) Vanderbilt Journal of Entertainment & Technology Law 515; Christophe Geiger and Oleksandr Bulayenko, 'Creating Statutory Remuneration Rights in Copyright law: What Policy Options under the International Framework?', in: Henning Grosse Ruse-Khan and Axel Metzger (eds.), Intellectual Property Ordering Beyond Borders (Cambridge University Press 2022) 408; Christophe Geiger, 'Empowering Remuneration Rights in Copyright law', Paper presented at the conference "Innovation, Justice, and Globalization – A Celebration of J.H. Reichman", Harvard Law School, Cambridge, USA, 27 September 2019 (on file with the author).

result of a balancing exercise that is inherent to any conflict between divergent fundamental rights. Despite some difficulties in reaching the optimal equilibrium, fundamental rights should remain the compass to navigate the seas of the still unexplored digital ecosystems and frontier technologies.[83] In this case, where the exploitation of copyright-protected works to train generative AI models challenged the anthropocentric nature and function of the copyright system, fundamental rights can be valid allies to reconcile the development of machine generated outputs by AI technology with human creativity, thus ensuring continued legitimacy as well as acceptance for the copyright system by creators and society at large.

Declaration of Competing Interest

The authors have no conflicts of interest to declare that are relevant to the content of the article titled "The Forgotten Creator: Towards a Statutory Remuneration Right for Machine Learning of Generative AI".

Data availability

No data was used for the research described in the article.

[83] On the numerous advantages of a fundamental rights framing of IP and the consequences to draw from it, see Christophe Geiger, Christophe Geiger, '"Constitutionalising" Intellectual Property Law? The Influence of Fundamental Rights on Intellectual Property in the European Union' (2006) International Review of Intellectual Property and Competition 371.